

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS EXTENSION 1

YEAR 11 EXAMINATION

TASK 2 OF 3

JULY 2004

Time allowed: 70 minutes

Instructions:

- Show all necessary working
- Start each question on a new page
- Full marks may not be awarded for careless or badly arranged work
- Non-programmable calculators may be used
- This paper must be handed in on top of your answer sheets
- Answers must be written in blue or black pen
- Answers must be arranged in order and stapled securely. No claims for missing pages will be allowed.

Name: _____

Class: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/8	/8	/10	/10	/9	/8	/53

Question 1

- (i) Sketch the parabola $y = x^2 + 5x$ on the number plane (2)
- (ii) Find the equation of the axis of symmetry. (1)
- (iii) Find the minimum value of the parabola. (1)
- (iv) Solve the inequality $x^2 + 5x \geq 0$ (1)
- (v) Using the formula $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ (3)

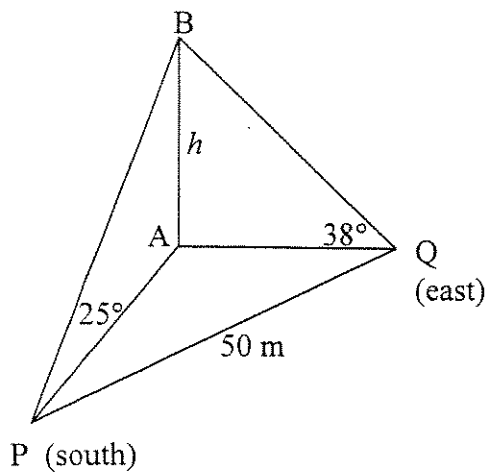
differentiate $y = x^2 + 5x$ from first principles.

Question 2

- (a) Differentiate the following with respect to x
 - (i) $y = x^3 - 3x^2 + 2x - 1$ (1)
 - (ii) $y = \frac{3x^3 + x^2}{x}$ (1)
 - (iii) $y = \frac{3x}{5x+1}$ (2)
 - (iv) $y = (6x-1)^4$ (2)
- (b) If $f(x) = 3x^2 + 2x - 1$
 - find (i) $f'(0)$ (1)
 - (ii) the gradient of the tangent at the point where $x = 1$. (1)

Question 3

- a) Simplify $\frac{\sin 2\theta}{\sin \theta}$ (1)
- b) Show that $\sin(\alpha - \beta)\cos \alpha - \cos(\alpha - \beta)\sin \alpha = -\sin \beta$ (2)
- c) Solve $(\tan \alpha + 1)(\tan \alpha + \sqrt{3}) = 0$ for $-180^\circ \leq \alpha \leq 180^\circ$ (3)
- d) From a point P due south of a vertical tower the angle of elevation of the top of the tower is 25° and from a point Q due east of the tower the angle of elevation is 38° . The distance from P to Q is 50 m. (4)



- (i) Show that $AP = h \tan 65^\circ$
- (ii) Express AQ in a similar way
- (iii) Hence find h, the height of the tower.
- Answer to the nearest metre.

Question 4

- (a) Write the equation of the line $\frac{y-7}{x+1} = \frac{4}{3}$ in general form. (2)
- (b) Write down the gradient and y-intercept of $\frac{x}{a} + \frac{y}{b} = c$ (2)
- (c) Find the perpendicular distance from the point (2, -1) to the line $4x - 3y + 7 = 0$ (2)
- (d) Find the equation of the line which is perpendicular to $y = 2x - 3$ and also passes through (2, -3). (2)
- (e) Find the coordinates of A if P (7, 8) divides the join of A (p, q) and B (9, 10) internally in the ratio 5:2. (2)

Question 5

(a) Differentiate

(i) $y = \sqrt{x}$ (1)

(ii) $y = x^2\sqrt{x}$ (1)

(b) (i) Find the gradient of the tangent to the curve $y = \frac{x^3}{x^2 + 4}$ at the point where $x = 2$. (2)

(ii) Hence find the gradient of the normal at this point. (1)

(iii) Find the equation of the normal. (2)

(c) Show by a sketch of $y = 2x^2 - 4x + 5$ that it is a positive definite function. (2)

Question 6

(a) Find the equation of the line through the point of intersection of $2x + y + 3 = 0$ (2)
and $2x - y - 5 = 0$ and also passing through the point $(1, 1)$.

(b) Find the acute angle between the two lines given in part (a). Answer (2)
to the nearest minute.

(c) Write $2\cos\theta + \sin\theta = 1$ in the form $R\sin(\theta + \alpha) = 1$. (3)
Hence solve $2\cos\theta + \sin\theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$.

(d) Find the value of p for which $x^2 - 2x + p = 0$ has equal roots. (1)

End of Paper

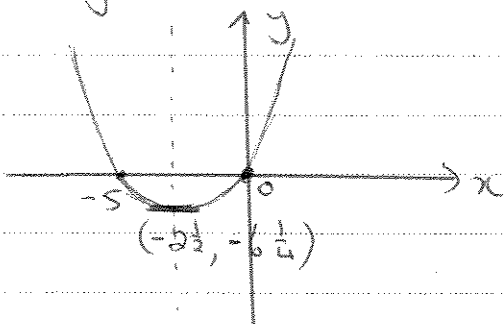
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Question 1

(i) $y = x^2 + 5x$

(1) (ii) $x = -2\frac{1}{2}$

(2)



(1) (iii) Min value
 $= -2\frac{1}{2}(-2\frac{1}{2} + 5)$
 $= -6\frac{1}{4}$

(1) (iv) $x^2 + 5x \geq 0 \Rightarrow x \geq 0$ and $x \leq -5$

(v) $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{[(x+h)^2 + 5(x+h)] - (x^2 + 5x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{[\cancel{x^2} + 2hx + h^2 + 5\cancel{x} + 5h - \cancel{x^2} - \cancel{5x}]}{h}$

(3)

$= \lim_{h \rightarrow 0} (2x + h + 5)$
 $= 2x + 5$

Question 2

a) (i) $y = x^3 - 3x^2 + 2x - 1$

$y' = 3x^2 - 6x + 2$

(ii) $y = \frac{3x^3 + x^2}{x}$

$= 3x^2 + x$

$y' = 6x + 1$

(iii) $y = \frac{3x}{5x+1}$

$y' = \frac{3(5x+1) - 5(3x)}{(5x+1)^2}$
 $= \frac{3}{(5x+1)^2}$

$$\begin{aligned}
 \text{(iv)} \quad y &= (6x-1)^4 \\
 y' &= 4(6x-1)^3 \times 6 \\
 &= 24(6x-1)^3
 \end{aligned}$$

$$\text{(b)} \quad f(x) = 3x^2 + 2x - 1$$

$$\text{(i)} \quad f'(x) = 6x + 2$$

$$\therefore f'(0) = 2$$

$$\text{(ii)} \quad f'(1) = 8$$

$\therefore$ grad. tangent at $x=1$ is 8.

Question 3

$$\text{a)} \quad \frac{\sin 2\theta}{\sin \theta} = \frac{2 \sin \theta \cos \theta}{\sin \theta} = 2 \cos \theta$$

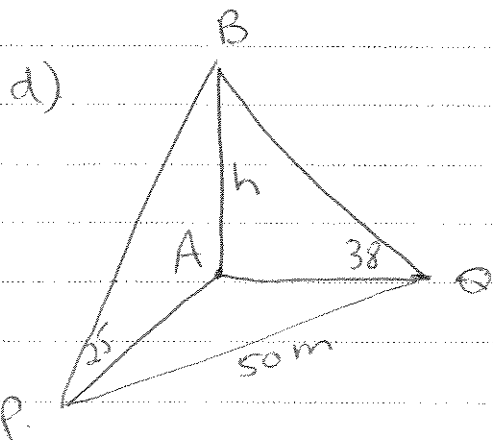
$$\begin{aligned}
 \text{b)} \quad \text{LHS} &= \sin(\alpha - \beta) \cos \alpha - \cos(\alpha - \beta) \sin \alpha \\
 &= \sin(\alpha - \beta - \alpha) \quad \text{using } \sin(A-B) \text{ result.} \\
 &= \sin(-\beta) \\
 &= -\sin \beta \quad (\text{4th quadrant result}) \\
 &= \text{RHS}
 \end{aligned}$$

$$\text{c)} \quad (\tan \alpha + 1)(\tan \alpha + \sqrt{3}) = 0$$

$$\Rightarrow \tan \alpha = -1 \quad \text{or} \quad \tan \alpha = -\sqrt{3}$$

$$\therefore \alpha = -45^\circ, 135^\circ, -60^\circ, 120^\circ$$

$$\text{ie } \alpha = -60^\circ, -45^\circ, 120^\circ, 135^\circ$$



$$\begin{aligned}
 \text{(i)} \quad \angle PBA &= 90 - 25 \\
 &= 65^\circ \quad (\triangle PBA \text{ right angled})
 \end{aligned}$$

$$\begin{aligned}
 \therefore \tan 65^\circ &= \frac{PA}{BA} \\
 &= \frac{PA}{h}
 \end{aligned}$$

$$\Rightarrow PA = h \tan 65^\circ$$

$$\begin{aligned}
 \text{(ii)} \quad \angle ABQ &= 90 - 38 \\
 &= 52^\circ
 \end{aligned}$$

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$$\therefore \tan 52^\circ = \frac{AQ}{AB} = \frac{AQ}{h}$$

$$\Rightarrow AQ = h \tan 52^\circ$$

(iii) By Pythagoras' theorem:

$$AP^2 + AQ^2 = PQ^2$$

$$h^2 \tan^2 65^\circ + h^2 \tan^2 52^\circ = 50^2$$

$$h^2 (\tan^2 65^\circ + \tan^2 52^\circ) = 2500$$

$$h^2 = \frac{2500}{\tan^2 65^\circ + \tan^2 52^\circ}$$

$$h = 20 \text{ m}$$

Question 4

$$a) \frac{y-7}{x+1} = \frac{4}{3}$$

$$3y - 21 = 4x + 4$$

$$\therefore 4x - 3y + 25 = 0$$

$$b) bx + ay = abc$$

$$\therefore y = \frac{-bx + abc}{a}$$

$$\text{grad} = -\frac{b}{a}$$

$$y \text{ intercept} = bc$$

$$c) d = \frac{|4 \times 2 - 3 \times -1 + 7|}{\sqrt{16 + 9}} = \frac{|8 + 3 + 7|}{5} = \frac{18}{5} = 3\frac{3}{5} \text{ units.}$$

$$d) y = 2x - 3 \text{ has gradient } 2$$

$$\therefore \text{gradient perpendicular} = -\frac{1}{2}$$

$$\therefore y - (-3) = -\frac{1}{2}(x - 2)$$

$$2y + 6 = -x + 2$$

$$x + 2y + 4 = 0$$

e) Using

$$x = \frac{nx_1 + mx_2}{m+n}$$

$$y = \frac{ny_1 + my_2}{m+n}$$

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$$7 = \frac{2p + 5 \times 9}{7}$$

$$\text{and } 8 = \frac{2q + 5 \times 10}{7}$$

$$49 = 2p + 45$$

$$56 = 2q + 50$$

$$2p = 4$$

$$2q = 6$$

$$p = 2$$

$$q = 3$$

Question 5

a) (i) $y = \sqrt{x} = x^{1/2}$
 $y' = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$

(ii) $y = x^2 \sqrt{x} = x^{5/2}$
 $y' = \frac{5}{2} x^{3/2}$
 $= \frac{5x\sqrt{x}}{2}$

b) (i) $y = \frac{x^3}{x^2+4}$
 $y' = \frac{3x^2(x^2+4) - 2x(x^3)}{(x^2+4)^2}$
 $= \frac{3x^4 + 12x^2 - 2x^4}{(x^2+4)^2}$
 $= \frac{x^4 + 12x^2}{(x^2+4)^2} = \frac{x^2(x^2+12)}{(x^2+4)^2}$

$\therefore$ gradient tangent at $x=2$
 is

$$m = \frac{2^2(2^2+12)}{(2^2+4)^2} = \frac{64}{64} = 1$$

(ii) gradient normal = -1.

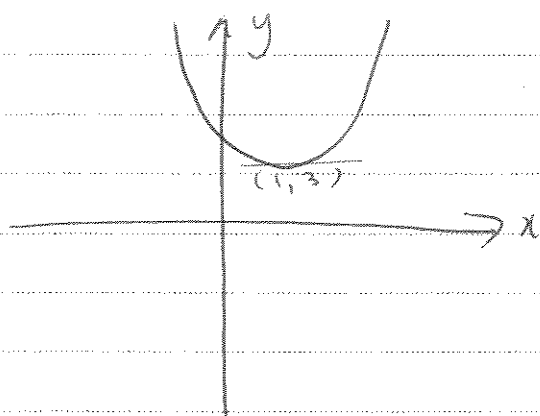
(iii) equation normal

$$y - 1 = -1(x - 2)$$

$$x + y - 3 = 0$$

c) $y = 2x^2 - 4x + 5$
 $= 2(x^2 - 2x) + 5$
 $= 2(x-1)^2 + 3$

Min. value of $(x-1)^2 = 0 \therefore$ Min. y value = 3
 when $x = 1$



$\therefore$ Parabola has no real roots and all y values positive $\Rightarrow$ positive definite

Question 6

a) $2x + y + 3 + k(2x - y - 5) = 0$

$(1, 1)$ satisfies

$$\therefore 2 + 1 + 3 + k(2 - 1 - 5) = 0$$

$$6 - 4k = 0 \Rightarrow k = \frac{3}{2}$$

$$\therefore 2x + y + 3 + \frac{3}{2}(2x - y - 5) = 0$$

$$4x + 2y + 6 + 6x - 3y - 15 = 0$$

$$10x - y - 9 = 0$$

OR

$$2x + y + 3 = 0 \quad (1)$$

$$2x - y - 5 = 0 \quad (2)$$

$$(1) + (2) \quad 4x - 2 = 0 \Rightarrow x = \frac{1}{2}$$

$$\text{In } (1) \quad 1 + y + 3 = 0 \Rightarrow y = -4$$

$$\therefore \frac{y-1}{x-1} = \frac{1-4}{\frac{1}{2}-1}$$

$$= \frac{5}{\left(\frac{1}{2}\right)} = 10$$

$$\therefore y-1 = 10x-10$$

$$10x - y - 9 = 0$$

(b) $2x + y + 3 = 0 \Rightarrow y = -2x - 3$

$$m_1 = -2$$

$$2x - y - 5 = 0 \Rightarrow y = 2x - 5$$

$$m_2 = 2$$

$$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

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$$\tan \theta = \left| \frac{-2-2}{1+(-2)(2)} \right|$$

$$= \left| \frac{-4}{-3} \right| = \frac{4}{3}$$

$$\therefore \theta (\text{acute}) = 53^\circ 8' \text{ (nearest minute)}$$

$$c) \quad 2\cos\theta + \sin\theta = 1$$

$$R = \sqrt{2^2 + 1^2}$$

$$= \sqrt{5}$$

$$\sqrt{5} \left(\frac{2}{\sqrt{5}} \cos\theta + \frac{1}{\sqrt{5}} \sin\theta \right) = 1$$

$$\sqrt{5} \sin(\theta + \alpha) = 1 \quad \text{where } \sin\alpha = \frac{2}{\sqrt{5}}$$

$$\cos\alpha = \frac{1}{\sqrt{5}} \Rightarrow \alpha = 63^\circ 26'$$

$$\therefore \sin(\theta + \alpha) = \frac{1}{\sqrt{5}}$$

$$\theta + \alpha = 26^\circ 34' \text{ or } 153^\circ 26' \text{ (1 revolution)}$$

$$\theta + 63^\circ 26' = 26^\circ 34' \text{ or } 153^\circ 26'$$

$$\therefore \theta = 26^\circ 34' - 63^\circ 26' \text{ or } 153^\circ 26' - 63^\circ 26'$$

$$= -36^\circ 52' \text{ or } 90^\circ$$

$$\therefore \text{For } 0^\circ \leq \theta \leq 360^\circ$$

$$\theta = 90^\circ \text{ or } 323^\circ 8'$$

$$d) \quad x^2 - 2x + p = 0$$

Equal roots when $\Delta = 0$.

$$\text{i.e. } b^2 - 4ac = 0$$

$$\therefore 4 - 4(1)(p) = 0$$

$$4 = 4p$$

$$\Rightarrow p = 1$$

